

Skills Summary

Comparing Fractions Without Being Fooled

Use this reference while completing your worksheet or when studying.

Skill 1 — Building an equivalent fraction

Rule: Multiply (or divide) the top *and* the bottom by the same number.

$$\frac{a}{b} = \frac{a \times k}{b \times k}$$

Multiplying by $\frac{k}{k}$ is multiplying by 1, so the amount does not change — the pieces are just re-cut. **Adding** the same number to top and bottom is a different fraction.

Example: Fill in: $\frac{2}{3} = \frac{\quad}{15}$

Step 1. The denominators tell me the scale factor, so I find what 3 was multiplied by and apply the same factor to the top.

Step 2. $3 \times 5 = 15$, so the numerator must also be multiplied by 5: 2×5 . Missing numerator: 10

Try it: Fill in: $\frac{3}{7} = \frac{\quad}{28}$

check: 12

Skill 2 — Comparing over a common denominator

Rule: Numerators can only be compared when the pieces are the same size.

Rewrite both fractions over a common denominator, then compare the counts.

Same numerator? The one with the *smaller* denominator is larger — fewer cuts means bigger pieces.

Example: Write < or > between $\frac{3}{4}$ and $\frac{5}{7}$.

Step 1. The pieces are different sizes, so comparing 3 against 5 would mean nothing — I rewrite both over a shared denominator first.

Step 2. $4 \times 7 = 28$: $\frac{3}{4} = \frac{21}{28}$ and $\frac{5}{7} = \frac{20}{28}$, and 21 twenty-eighths beats 20. $\frac{3}{4} > \frac{5}{7}$

Try it: Write < or > between $\frac{2}{9}$ and $\frac{3}{8}$.

check: $\frac{8}{3} > \frac{6}{2}$

Skill 3 — Using one half as a benchmark

Rule: A fraction is more than one half when its numerator is more than half of its denominator.

If one fraction is below one half and the other is above it, the comparison is finished — no common denominator needed. If both land on the same side of the benchmark, fall back to Skill 2.

Example: Write $<$ or $>$ between $\frac{4}{9}$ and $\frac{5}{8}$.

Step 1. Before doing any arithmetic I check each fraction against one half, because a split verdict decides the comparison on its own.

Step 2. Half of 9 is 4.5 and 4 is below it, so the first is under a half; half of 8 is 4 and 5 is above it, so the second is over a half. $\frac{4}{9} < \frac{5}{8}$

Try it: Write $<$ or $>$ between $\frac{7}{15}$ and $\frac{9}{16}$.

check: $\frac{9}{16} > \frac{7}{15}$

(!) Watch out: Two habits that look like reasoning and are not. Comparing only the tops (“7 beats 5, so seven twelfths beats five sixths”) ignores piece size. Comparing only the bottoms (“8 beats 5, so three eighths beats three fifths”) gets the order exactly backwards.